VOL. I · NO. 2 MONDAYS & THURSDAYS

THE LAUNCH REPORT

Rocket launches, reentries and the business of spaceflight, worldwide

THURSDAY, OCTOBER 15, 2026 ALL TIMES EASTERN DAYLIGHT TIME (EDT)

THE WEEK IN LAUNCH

Three launches since Monday, Dragon arrives today, and a Space Force flight tonight

SpaceX launched twice in ten hours, sending 27 Starlink satellites up from California on Monday night and NASA's CRS-35 cargo Dragon toward the International Space Station (ISS) on Tuesday morning. On Wednesday, China's LandSpace put ten satellites in orbit on a methane-fueled Zhuque-2E, eight of them for an orbital computing constellation. That brings 2026 to 242 orbital launch attempts, according to GCAT.

Dragon is due to dock at about 11 a.m. EDT today, and a Falcon 9 launches the classified USSF-481 mission from Vandenberg at 7:44 p.m. EDT. Japan's H3 carries the Martian Moons eXploration (MMX) probe on Monday. The Federal Aviation Administration (FAA) opened an environmental review of up to 456 Starship launches a year from Starbase, Texas, and ISRO's NVS-03 window opens today with no launch date announced.

I · RECENT LAUNCHES & RETURNS

Three launches, three successes

Two Falcon 9s, including the first Starlink flight in more than three weeks, and a Chinese commercial rocket.



A Falcon 9 lifts off from Space Launch Complex 4E at Vandenberg Space Force Base with NASA's SWOT satellite on Dec. 16, 2022. Monday's Starlink 15-25 flew from the same pad.

PHOTO: NASA/KEEGAN BARBER · PUBLIC DOMAIN

Falcon 9: Starlink 15-25

After four date changes, Falcon 9 lifted off from Space Launch Complex 4E at Vandenberg Space Force Base, California, at 8:28 p.m. EDT Monday (00:28 UTC Tuesday) with 27 Starlink V2 Mini satellites, Spaceflight Now reported. Booster B1088 landed on the drone ship Of Course I Still Love You on its 20th flight. It was the first Falcon 9 Starlink launch since September 19, Space.com reported.

Falcon 9: CRS-35

Falcon 9 launched NASA's 35th SpaceX cargo mission from Space Launch Complex 40 at Cape Canaveral at 6:33 a.m. EDT Tuesday, and booster B1096 landed at Landing Zone 40 on its eighth flight, SpaceX posted. Cargo Dragon C211 carries more than 6,300 pounds, including the last pair of ISS Roll-Out Solar Arrays (NASA).

Zhuque-2E: ten satellites

LandSpace's Zhuque-2E Y4 lifted off from the Dongfeng commercial space innovation pilot zone at Jiuquan at 7:40 p.m. Beijing time Wednesday, 7:40 a.m. EDT, and deployed 10 satellites into their preset orbits, Xinhua reported. Eight belong to ADA Space's "star compute" constellation, planned to reach 2,800 computing satellites, and fly in a dawn-dusk orbit; two are LandTower laser-communication test satellites. It was the 10th flight of the Zhuque-2 family, LandSpace said.

USSF-481 tonight, then H3 to Mars

Every orbital launch with a date between now and October 29, plus missions targeting later this month. Times are EDT, with Coordinated Universal Time (UTC) below; NET means no earlier than.

DATE / TIME	VEHICLE	PROVIDER	MISSION / PAYLOAD	SITE	ORBIT	STATUS	SOURCES
Thu Oct 15, 7:44 p.m. 23:44 UTC	Falcon 9	SpaceX for the U.S. Space Force	USSF-481 (TH-2). Booster B1081, 28th flight	SLC-4E, Vandenberg, Calif.	Polar LEO	CONFIRMED	FAA; Spaceflight Now
Oct 15–20 target	GSLV Mk II	Indian Space Research Organisation (ISRO)	NVS-03 (IRNSS-1L) navigation satellite	Second Launch Pad, Sriharikota, India	GTO	PRELIMINARY	Times of India
Mon Oct 19, 6:30 a.m. 10:30 UTC	Falcon 9	SpaceX	Bandwagon-5 mid-inclination rideshare, 31 separation events. Booster B1085, 19th flight	SLC-40, Cape Canaveral, Fla.	Mid-inclination LEO	CONFIRMED	FAA; Spaceflight Now
Mon Oct 19, 3:41:03 p.m. 19:41:03 UTC	H3-24	Mitsubishi Heavy Industries (MHI) for JAXA	Martian Moons eXploration (MMX), Phobos sample-return probe	LP-2, Yoshinobu complex, Tanegashima, Japan	Mars transfer	CONFIRMED	JAXA launch plan; Spaceflight Now
Tue Oct 20, 6:14 p.m. 22:14 UTC	Falcon 9	SpaceX	Starlink 15-15: 27 Starlink V2 Mini satellites. Booster B1063, 36th flight	SLC-4E, Vandenberg, Calif.	LEO	CONFIRMED	FAA; Spaceflight Now
NET Oct 21	Electron	Rocket Lab	"We're Not Hawking Around": three HawkEye 360 satellites, 98th Electron	Launch Complex 1, Mahia, New Zealand	LEO, 520 km, 97°	PRELIMINARY	Spaceflight Now
Sat Oct 24, 1:29 a.m. 05:29 UTC	Falcon 9	SpaceX	Starlink 15-16: 27 Starlink V2 Mini satellites	SLC-4E, Vandenberg, Calif.	LEO	TENTATIVE	FAA; Launch Library 2
NET Oct 29	Vulcan VC6L	United Launch Alliance (ULA)	Amazon Leo LV-01 broadband satellites	SLC-41, Cape Canaveral, Fla.	LEO	PRELIMINARY	Launch Library 2
NET October	Long March 12A	CASC	Flight 2	Jiuquan, China	LEO	PRELIMINARY	Launch Library 2
NET October	SSLV	ISRO	AayulSAT-1A and others	First Launch Pad, Sriharikota, India	LEO	PRELIMINARY	Launch Library 2

CONFIRMED date and time announced by the launch provider, its customer or the FAA. **TENTATIVE** specific date in published schedules or airspace and maritime notices, not yet confirmed by the operator. **PRELIMINARY** target range, or no-earlier-than date or month only.

Abbreviations: SLC, Space Launch Complex; SLC-4E, Space Launch Complex 4 East; LP-2, Launch Pad 2; LEO, low Earth orbit; GTO, geostationary transfer orbit; GSLV, Geosynchronous Satellite Launch Vehicle; SSLV, Small Satellite Launch Vehicle; CASC, China Aerospace Science and Technology Corp.; JAXA, Japan Aerospace Exploration Agency.

Date changes

Starlink 15-25. Listed Monday for 8:11 p.m.; flew at 8:28 p.m. EDT Monday. No reason given (Spaceflight Now).

Bandwagon-5. NET October on Monday; now dated October 19 at 6:30 a.m. with an FAA airspace window. No reason given (Spaceflight Now).

Ariane 64 / Amazon Leo LE-04. NET October on Monday; now NET November 23 in Launch Library 2. No reason given.

NVS-03. The October 15–20 window opens today without an ISRO date. Engineers are studying whether to fly the satellite with a pyro valve open to avoid a repeat of the NVS-02 failure, The Federal reported.

What's flying

New on the schedule. Starlink 15-15 and 15-16 from Vandenberg and Rocket Lab's 98th Electron, for HawkEye 360, have been added since Monday (Spaceflight Now).

H3 / MMX. Liftoff is 4:41:03 a.m. Japan Standard Time on Tuesday, October 20; the reserve period runs to November 7 (JAXA launch plan). MMX is to return Phobos samples to Earth in fiscal 2031, Spaceflight Now notes.

QUICK HIT

Starship. NASASpaceflight's unofficial schedule still shows Flight 15 NET November 8 from Starbase Pad 2. Ship 40, the first Starship to return to its launch site after flying, now stands in a Megabay, SpaceX posted.



Launch sites with orbital launches scheduled through November 8; numbers count launches. Filled: dated launch (Confirmed or Tentative). Hollow: Preliminary. Vandenberg and Cape Canaveral flew on October 12 and 13; Jiuquan on October 14.

MAP: THE LAUNCH REPORT · PAD COORDINATES FROM LAUNCH LIBRARY 2 · NATURAL EARTH

III · TRAJECTORIES & HAZARD AREAS

Where H3's spent stages will fall

JAXA has published the sea areas where MMX's boosters, nose fairing and first stage are expected to come down.

H3 Flight 10 timeline

According to the launch plan, the rocket turns its pitch plane to an azimuth of 90 degrees, due east, shortly after liftoff and flies out over the Pacific. The upper stage fires twice and releases MMX into a Mars transfer orbit 71 minutes after liftoff. The times below assume launch at 4:41:03 a.m. JST on October 20 (3:41:03 p.m. EDT October 19). Because the probe is headed to Mars, events after the first second-stage cutoff shift with the launch date.

EVENT	TIME AFTER LIFTOFF	ALTITUDE
Solid rocket boosters separate (two pairs)	1:56 / 1:58	59–61 km
Payload fairing separates	3:17	131 km
Main engine cutoff (MECO)	4:52	208 km
First and second stages separate	4:59	214 km
Second-stage engine start 1 (SELI1)	5:12	224 km
Second-stage engine cutoff 1 (SECO1)	10:40	329 km
Second-stage engine start 2 (SELI2)	1:03:57	322 km
Second-stage engine cutoff 2 (SECO2)	1:10:21	497 km
MMX separates, 11.2 km/s	**1:11:12**	**592 km**

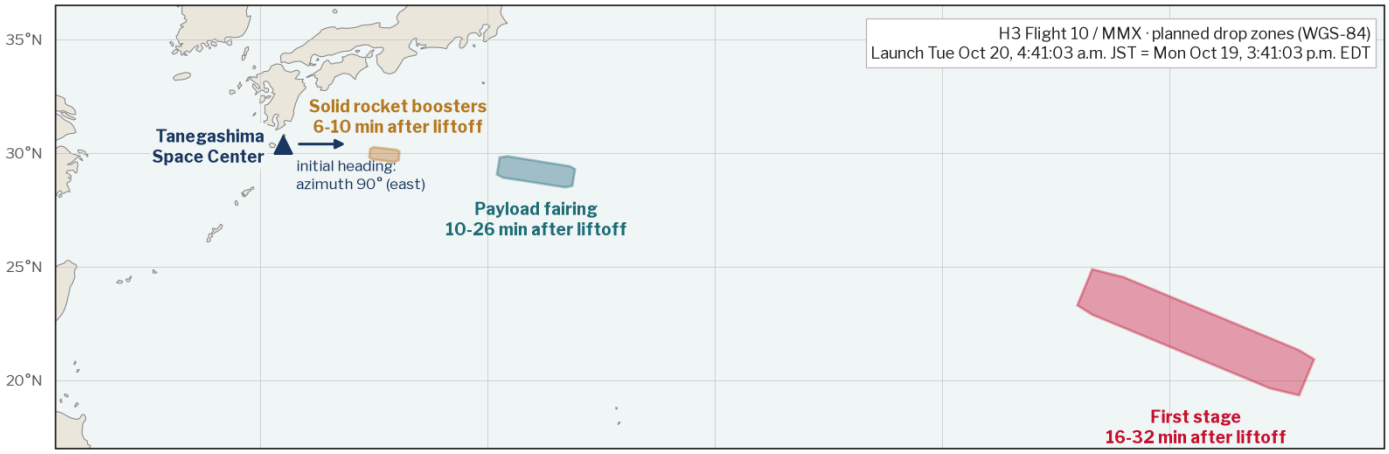


JAXA's HTV-X1 cargo spacecraft approaches the International Space Station on Oct. 29, 2025, four days after launching from Tanegashima on an H3-24, the same H3 version that will carry MMX.

PHOTO: NASA · PUBLIC DOMAIN

Tonight's Vandenberg window

The FAA's air traffic operations plan lists SpaceX's TH-2 mission, the USSF-481 launch, from Vandenberg with a primary window of 7:44 p.m. to 12:27 a.m. EDT (23:44–04:27 UTC) tonight and a backup from 7:35 p.m. Friday. It also lists Bandwagon-5 from Cape Canaveral from 6:30 to 8:13 a.m. EDT on October 19 and Starlink 15-15 from Vandenberg from 6:14 to 10:57 p.m. EDT on October 20 (FAA advisory). USSF-481 is headed for a polar orbit, according to Launch Library 2. No hazard-area coordinates have been published for these flights.



Planned drop zones for H3 Flight 10, plotted from the corner coordinates in JAXA's launch plan (World Geodetic System 1984). The solid rocket booster zone measures 62 by 128 km, the fairing zone 107 by 330 km and the first-stage zone 226 by 1,105 km. Ships and aircraft are warned to stay out of these zones through Japan's coast guard and transport ministry. The arrow shows the eastward heading the plan gives for shortly after liftoff; it is not a ground track.

MAP: THE LAUNCH REPORT · DATA FROM THE JAXA H3/MMX LAUNCH PLAN (AUG. 20, 2026) · NATURAL EARTH

IV · REENTRIES & ISS TRAFFIC

India's SpaDeX pair comes down; Dragon docks today

Six objects reentered on October 12 and 13; an Indian rocket stage and a Russian satellite are next.

Confirmed reentries

CelesTrak's satellite catalog lists these objects as decayed. They include both satellites of ISRO's Space Docking Experiment (SpaDeX), launched December 30, 2024.

OBJECT (CATALOG NO.)	LAUNCHED	DECAYED	LAST ORBIT, KM
SDX01, SpaDeX (62459)	**Dec 30, 2024**	**Oct 12**	**148 × 157**
SDX02, SpaDeX (62460)	**Dec 30, 2024**	**Oct 12**	**152 × 161**
Kuaizhou-11 stage (100743)	Sep 17, 2026	Oct 12	110 × 280
Starlink-1337 (45582)	Apr 22, 2020	Oct 12	148 × 159
Electron kick stage (64230)	Jun 2, 2025	Oct 13	171 × 176
Starlink-1810 (46709)	Oct 18, 2020	Oct 13	169 × 172

Predicted

A Polar Satellite Launch Vehicle (PSLV) upper stage from the February 2022 EOS-04 launch was predicted to reenter at 11:16 p.m. EDT Wednesday, plus or minus 13 hours, and Russia's Cosmos 2572 satellite at 12:46 a.m. EDT Sunday, October 18, plus or minus 28 hours (Aerospace). Both are in near-polar orbits, so they could come down almost anywhere.



One orbit of each object, computed from CelesTrak's latest orbital elements (October 15 for both). Reentry can occur anywhere along later orbits, which shift westward each revolution.

MAP: THE LAUNCH REPORT · CELESTRAK · NATURAL EARTH



A SpaceX cargo Dragon backs away from the International Space Station after undocking from the Harmony module's forward port on June 16, 2026, above a cloudy North Pacific.

PHOTO: NASA · PUBLIC DOMAIN

Station traffic

CRS-35 is due to dock on its own at the space-facing port of Harmony at about 11 a.m. EDT today, with NASA coverage from 9:15 a.m. EDT (NASA). It is to stay about a month. The next Russian cargo ship, Progress MS-36, is listed for 12:03 a.m. Eastern Standard Time (EST) on November 24 from Baikonur (Launch Library 2).

FAA opens a review of 456 Starship launches a year in Texas

A new environmental study for Starbase joins the Florida reviews and the Part 450 rule changes now out for comment.

Starbase

The FAA will prepare a Programmatic Environmental Impact Statement (PEIS) on SpaceX's proposal to increase Starship-Super Heavy operations at the Boca Chica launch site at Starbase and to add Starship landing and expenditure locations, according to a notice published October 14 (Federal Register, docket FAA-2026-12808). SpaceX proposes up to 100 launches and 200 landings a year at first, then 456 launches and 912 landings, against 25 launches allowed now, Invezz reported. In-person scoping meetings are set for November 17 and 19, a virtual one for November 18, and comments are due November 30 (SpacePolicyOnline).

Florida and Part 450

Starship at SLC-37. Scoping comments on up to 76 Starship launches a year at Cape Canaveral closed Wednesday (notice, docket FAA-2026-10792).

New Glenn at SLC-36. A virtual public meeting on raising New Glenn's cap from 12 to 50 launches a year is set for October 22 from 6 to 8 p.m. EDT; comments close October 30 (notice, docket FAA-2026-9841).

Part 450. Comments on five proposed changes to the licensing rule for U.S. commercial launches and reentries are due November 4, including fully digital license applications (docket FAA-2026-9935) and simpler flight safety analysis (FAA-2026-9936).

Ship 40 back home

Ship 40, which made Starship's first intact ocean splashdown on Flight 13 and was towed to Christmas Island after 24 days adrift, now stands upright in a Megabay at Starbase. SpaceX says it "will provide first-of-its-kind data on a flown Starship heatshield, Raptor engines, avionics, and more" (SpaceX; TeslaNorth).

QUICK HIT

Orbital computing. ADA Space's second group of "star compute" satellites, launched Wednesday, will test high-performance computing payloads, laser links and large radiator wings, China Daily reported; a third group is planned for 2026.

Starlink and Ukraine

Ukrainian officials said military Starshield terminals were restricted while moving at 72 to 161 km/h overnight into October 14. Elon Musk said limits on Starshield "are decided by the US military and ultimately POTUS," Politico reported.

242 orbital launch attempts so far in 2026

SpaceX accounts for almost half. China needs 28 more launches to pass 100 in a year for the first time.

242	235	118	72	28
orbital launch attempts worldwide, Jan 1–Oct 14	reached orbit	by SpaceX, all successful	by China, four failures	more Chinese launches needed this year to pass 100

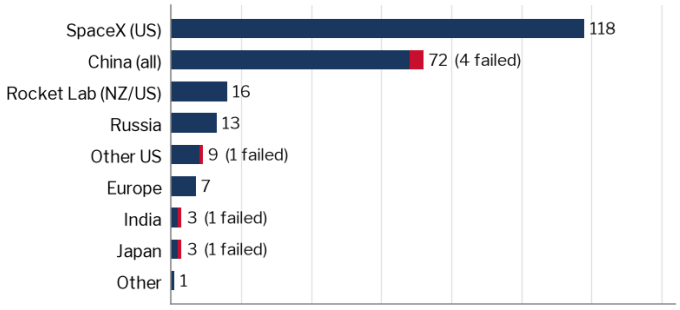
The count

Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) records 242 orbital launch attempts through October 14, 235 of them successful, including Wednesday's Zhuque-2E. SpaceX has flown 118, all successful, followed by China with 72, Rocket Lab with 16 and Russia with 13. Nine orbital launches have flown so far in October.

Busiest launch sites

Vandenberg Space Force Base in California remains the year's busiest launch site, ahead of Florida's Cape Canaveral and Kennedy Space Center combined (GCAT, through October 14).

SITE	LAUNCHES, 2026
Vandenberg Space Force Base, California	**65**
Cape Canaveral and Kennedy Space Center, Florida	57
Jiuquan, China	27
Mahia, New Zealand (Rocket Lab)	15
Hainan commercial launch site, Wenchang, China	14
Taiyuan, China	12
Plesetsk, Russia	8



Orbital launch attempts in 2026 by provider or country. Navy: success. Red: failure or partial failure.

CHART: THE LAUNCH REPORT · GCAT (JONATHAN MCDOWELL) THROUGH OCT 14

The failures

According to GCAT, seven launches have failed this year: India's PSLV on January 12; China's Long March 3B on January 16, Ceres-2 on January 17, Tianlong-3 on April 3 and Long March 7A on August 10; Japan's Kairos on March 5; and Blue Origin's New Glenn on April 19, recorded as a partial failure.

October so far

All nine orbital launches this month have succeeded; six were SpaceX Falcons (Launch Library 2).

DATE, EDT	ROCKET	MISSION	SITE
Oct 1, 11:10 a.m.	Falcon 9	Crew-13	Cape Canaveral
Oct 1, 2:32 p.m.	Falcon 9	Transporter-18 rideshare	Vandenberg
Oct 1, 11:54 p.m.	Falcon Heavy	NROL-97	Kennedy
Oct 6, 11:25 p.m.	Nuri	NeonSat-2 to 6	Naro, South Korea
Oct 9, 3:27 p.m.	Long March 12	SatNet LEO Group 27	Wenchang, China
Oct 10, 3:39 a.m.	Falcon 9	SDA Tranche 1 Transport Layer A	Vandenberg
Oct 12, 8:28 p.m.	Falcon 9	Starlink 15-25	Vandenberg
Oct 13, 6:33 a.m.	Falcon 9	CRS-35	Cape Canaveral
Oct 14, 7:40 a.m.	Zhuque-2E	10 satellites	Jiuquan, China

Takeaways

1. Three launches since Monday: Starlink 15-25, CRS-35 and a Zhuque-2E with ten satellites.
2. CRS-35 docks at about 11 a.m. EDT today; USSF-481 launches at 7:44 p.m. EDT.
3. H3 launches Japan's MMX toward Mars at 3:41 p.m. EDT Monday.
4. The FAA opened an environmental review of up to 456 Starship launches a year at Starbase.
5. Both SpaDeX satellites reentered on October 12.

KEY DATES AHEAD (EDT)

Thu Oct 15	CRS-35 docks, about 11 a.m.; USSF-481, 7:44 p.m.
Sun Oct 18	Cosmos 2572 reentry predicted, about 12:46 a.m.
Mon Oct 19	Bandwagon-5, 6:30 a.m.; H3 / MMX, 3:41 p.m.
Tue Oct 20	Starlink 15-15, 6:14 p.m.
Thu Oct 22	New Glenn cadence meeting, 6–8 p.m.
NET Oct 29	Vulcan returns with Amazon Leo
Nov 17–19	Starbase scoping meetings; comments due Nov 30

SOURCES

Schedules: Launch Library 2 (The Space Devs), Spaceflight Now, NASASpaceflight, FAA operations plan advisory. Agencies: NASA, JAXA, FAA, Federal Register. Tracking: CelesTrak, The Aerospace Corporation, GCAT (Jonathan McDowell). Reporting: Xinhua, LandSpace, China Daily, Space.com, Times of India, The Federal, Invezz, SpacePolicyOnline, TeslaNorth, Politico. X posts: @SpaceX. Photos: NASA (public domain). Maps: Natural Earth (public domain).

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